MIL-D-13791E
26 June 1981
SUPERSEDING
MIL-D-13791D
30 December 1966

MILITARY SPECIFICATION

DISTRIBUTOR, IGNITION SYSTEM: INTEGRAL COIL, WATERPROOF, 24-VOLT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers waterproof ignition distributors, with integral ignition coil, of breaker point or breakerless design, for operation in 24 volt direct current (Vdc) electrical systems. Specification also covers authorized repair parts (see 3.1.2).

1.2 Classification. Distributors covered by this specification shall be classified in the following manner (see 6.2):

Type I	- Left Hand (counterclockwise) rotation viewing drive end.
Type II	- Right Hand (clockwise) rotation viewing drive end.
Class 1	- Breaker point design.
Class 2	- Breakerless design.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2920

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

MIL-F-13927 - Fungus Resistance Test; Automotive Components.

STANDARDS

FEDERAL

FED-STD-H28/2 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461 - Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1184 - Electrical Components for Automotive Vehicles, Waterproofness Tests.

DRAWINGS

ARMY

5346744 - Ignition Distributor Contact Set.
5701239 - Ignition Distributor Contact Set.
7059538 - Ignition Distributor Contact Set.
7348578 - Ignition Distributor Rotor.
7353276 - Ignition Distributor Assembly.
7355596 - Ignition Distributor Assembly.
7355734 - Ignition Distributor Assembly.
7358569 - Ignition Distributor Assembly.
7374880 - Ignition Distributor Cap.
7374883 - Ignition Distributor Cap Cover.
7375373 - Ignition Distributor Cap Cover.
7375377 - Ignition Distributor Cap.
7539588 - Ignition Distributor Rotor.
7727531 - Gap, Sparking.
7762671 - Ignition Distributor Assembly.
7762685 - Ignition Distributor Assembly.
8360076 - Ignition Distributor Assembly.
8712398 - Ignition Distributor Assembly.
8722526 - Ignition Distributor Cap.
8722527 - Ignition Distributor Rotor.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

8737853 - Ignition Distributor Contact Set.
8754337 - Adaptor, Distributor Mounting.
10863469 - Ignition Distributor Assembly.
11660529 - Ignition Distributor Assembly.
11663066 - Ignition Distributor Coil.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification requirements.

3.1.1 Distributor assemblies. Distributor assemblies furnished under this specification shall be products which are qualified for listing on applicable Qualified Products List (QPL) at the time set for opening of bids (see 6.3).

3.1.2 Authorized repair parts. Repair parts furnished under this specification shall be only those specified for authorized repairs (see table I), and shall be products which are qualified (see 3.1.2.1 and 3.1.2.2) for listing on applicable QPL at the time set for opening of bids (see 6.3).

3.1.2.1 Assembled repair parts. Repair parts requiring qualification to this specification will be automatically qualified when those parts have been contained in a distributor assembly that has been qualified.

3.1.2.2 Separate repair parts. Repair parts submitted separately for qualification shall be qualified as follows: Repair part shall be assembled with all other parts necessary to form a complete distributor assembly. Repair part will be considered qualified when distributor assembly containing part has been qualified.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free of defects which adversely affect performance or serviceability of the finished product (see 6.7).

3.3 Design and construction. Distributor assemblies and authorized repair parts shall conform to the form and dimensions specified on drawings listed in table I, or other applicable drawings or MS standards (see 6.2).

TABLE I. Distributor assemblies and authorized repair parts.

Drawing number	Description
7355734	Type I - Ignition Distributor Assembly.
7762671	Type I - Ignition Distributor Assembly.
7762685	Type I - Ignition Distributor Assembly.
8360076	Type I - Ignition Distributor Assembly.
8712398	Type I - Ignition Distributor Assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

TABLE I. Distributor assemblies and authorized repair parts. - Continued

Drawing number	Description	
11660529	Type I -	Ignition Distributor Assembly.
7353276	Type II -	Ignition Distributor Assembly.
7355596	Type II -	Ignition Distributor Assembly.
7358569	Type II -	Ignition Distributor Assembly.
10863469	Type II -	Ignition Distributor Assembly.
7374880	ARP 1/ -	Ignition Distributor Cap.
7375377	ARP -	Ignition Distributor Cap.
8722526	ARP -	Ignition Distributor Cap.
7374883	ARP -	Ignition Distributor Cap Cover.
7375373	ARP -	Ignition Distributor Cap Cover.
11663066	ARP -	Ignition Distributor Coil.
5346744	ARP -	Ignition Distributor Contact Set.
5701239	ARP -	Ignition Distributor Contact Set.
7059538	ARP -	Ignition Distributor Contact Set.
8737853	ARP -	Ignition Distributor Contact Set.
7348578	ARP -	Ignition Distributor Rotor.
7539588	ARP -	Ignition Distributor Rotor.
8722527	ARP -	Ignition Distributor Rotor.

1/ ARP = Authorized repair part.

3.3.1 Interchangeability. All distributor assemblies and authorized repair parts shall be physically and functionally interchangeable with all other products of the same part number.

3.3.2 Housing. Distributor housing shall be so constructed as to be electrically continuous.

3.3.2.1 Coil-to-distributor lead. When specified (see 6.2), provision shall be made to permit connection to coil-to-distributor lead for testing purposes.

3.3.3 Timing: Initial setting. Timing of distributor shall be accomplished by rotation of complete distributor assembly in its mounting.

3.3.3.1 Advance and retard. Spark advance mechanism shall be centrifugal.

3.3.4 Drive arrangement. Distributor shaft speed shall be one-half crankshaft speed. Drive end of the shaft shall be designed to prevent assembly 180 degrees out of line.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

3.3.5 Ventilation. Provision shall be made for ventilation as specified on applicable drawing or MS standard.

3.3.6 Lubrication. Breaker cam follower shall be so designed as to require no additional lubrication prior to 100 hours of operation. Rotor shaft bearing for class 1 distributor shall be so designed as to require no additional lubrication prior to 600 hours of operation, and 1500 hours for class 2 distributor.

3.3.7 Threaded parts. Screw threads of the form, number per inch, and class specified on the applicable drawing or military standard shall be in accordance with FED-STD-H28/2.

3.4 Performance.

3.4.1 Voltage output. Distributor assembly shall produce minimum voltage output at each output terminal under operating conditions specified in table II. Variation of output voltage from output terminal to output terminal shall be no more than 10 percent. In no event shall operating distributor assembly produce less than 6000 Vdc at output terminals.

TABLE II. Voltage output.

Distributor shaft speed (rpm)	Voltage output, minimum (Kilovolts)			
	Ambient air temperature 77° ± 15°F		Ambient air temperature 200° ± 10°F	
	18V input	30V input	18V input	30V input
300 ± 10	14	17	10	15
1800 ± 25	12	17	10	14

3.4.1.1 Low input voltage (class 2). Class 2 distributor assembly shall withstand operation at any steady state system input voltage from zero to 18 Vdc, of correct polarity, without impairing performance when operated as specified in 3.4.1.

3.4.1.2 High input voltage (class 2). Class 2 distributor assembly shall withstand operation for one hour at steady state system input voltage of not less than 30 Vdc and not more than 40 Vdc, of correct polarity, without impairing performance when operated as specified in 3.4.1.

3.4.2 Timing performance. Distributor assembly shall provide voltage output, at each output terminal in turn, in regular rotary sequence. Peak voltage output shall occur within ± 1 degree of rotation for each point of actuation (see 6.5.1) on distributor shaft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

3.4.2.1 Advance and retard performance. Centrifugal spark advance mechanism (see 3.3.3.1) shall continuously adjust timing of voltage output at spark gap in conformance to advance curve specified on applicable drawing or MS standard.

3.4.3 Dielectric strength. Insulation materials shall exhibit no loosening, cracking, charring, burning, smoking, nor reduction in dielectric strength after distributor assembly operation, with two output leads connected to ground and remaining leads connected to spark gap set at eight millimeters, with 32 Vdc input. Subsequently, distributor assembly shall conform to 3.4.1.

3.4.4 Endurance.

3.4.4.1 Class 1 distributor assembly. Class 1 distributor assembly, including distributor shaft to crankshaft coupling, shall operate as in intended use for 600 hours with no maintenance except resetting of breaker points at 100 hour intervals, and lubrication of breaker cam, breaker lever pivot and felt wick under rotor. Subsequently, distributor assembly shall operate as specified in 3.4.1 and 3.4.2.

3.4.4.2 Class 2 distributor assembly. Class 2 distributor assembly, including distributor shaft to crankshaft coupling, shall operate as in intended use for 1500 hours with no maintenance. Subsequently, distributor assembly shall operate as specified in 3.4.1 and 3.4.2.

3.4.5 Polarity, class 2 distributor. Class 2 distributor assembly shall operate in negatively grounded system, and shall be inoperable during reversed polarity. Subsequent to applications of reversed polarity, distributor assembly shall operate as specified in 3.4.1.

3.4.6 Transient voltage characteristics. Distributor assembly shall undergo input of correctly polarized 240 Vdc transient in 200°F ambient air temperature. Subsequently, distributor assembly shall operate as specified in 3.4.1.

3.4.7 Waterproofness. Distributor assembly shall undergo, and shall pass, waterproofness test procedure specified for type I, class 1 components in MIL-STD-1184.

3.4.8 Corrosion resistance. Distributor assembly shall undergo test procedure in accordance with method 101 of MIL-STD-202, except the test period shall be 200 hours. Subsequently, distributor assembly shall operate as specified in 3.4.1 and conform to 3.5.

3.4.9 Fungus resistance. Distributor assembly shall undergo test procedure in accordance with method B of MIL-F-13927. Subsequently, distributor shall operate as specified in 3.4.1 and conform to 3.5.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

3.4.10 Vibration resistance. Distributor assembly shall undergo test procedure in accordance with method 201 of MIL-STD-202. Assembly shall operate as in intended use during test procedure. Subsequently, assembly shall operate as specified in 3.4.1.

3.4.10.1 Distributor with pitch- or oil-filled coil. Distributor assembly with pitch- or oil-filled coil shall withstand 72 hours of vibration as specified in 3.4.10, and subsequently shall operate as specified in 3.4.1.

3.4.10.2 Distributor with molded or potted coil. Distributor assembly with molded or potted coil shall withstand 96 hours of vibration as specified in 3.4.10, and subsequently shall operate as specified in 3.4.1.

3.4.10.3 Coil mounting. Coil mounting tabs, screws, and clamp shall not loosen, crack, or break during eight hours of vibration at 20g input while retaining coil as in intended use.

3.4.11 Shock resistance. Distributor assembly shall undergo test procedure in accordance with condition A, method 213 of MIL-STD-202. Subsequently, assembly shall operate as specified in 3.4.1.

3.4.12 Electromagnetic interference characteristics. Distributor assembly shall conform to requirements of classes A3 and C1 of MIL-STD-461 for tactical vehicle components.

3.4.12.1 Resistor-suppressors. Resistor-suppressors, in output towers of distributor cap, shall provide the following resistance to high peak voltage pulses (see 3.4.4):

Initial value (ohms)	Value (ohms) after cycling
3750-6250	3750-7500

3.4.13 High temperature operation. Distributor assembly shall operate as specified in 3.4.1 at 200°F ± 10°F ambient air temperature.

3.4.14 Low temperature operation. Distributor assembly shall operate as specified in 3.4.1 after distributor assembly temperature has been stabilized at minus 65°F ± 5°F.

3.4.15 High temperature stall. Distributor assembly shall operate as specified in 3.4.1, and shall exhibit no damage to coil or ballast resistor, after undergoing 6 hours with 24 Vdc applied to coil primary in an ambient air temperature of 160°F ± 5°F.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

3.4.16 Low temperature stall. Distributor assembly shall operate as specified in 3.4.1, and shall exhibit no damage to coil or ballast resistor, after undergoing 6 hours with 24 Vdc applied to coil primary, in an ambient air temperature of minus 65°F ± 5°F.

3.5 Finish. Distributor shall exhibit no blistering, chipping, peeling, or rust after undergoing tests.

3.6 Marking. A nameplate conforming to type I of MIL-P-514 shall be affixed to distributor housing, as shown on applicable drawing, and shall include following minimum information:

Ignition system
24 Volts
Military part number
US
National stock number
Manufacturer's part number
Manufacturer's serial number
Manufacturer's identification (name or code)
Direction of rotation (see 3.6.1)

3.6.1 Direction of rotation. Direction of rotation shall be indicated by an arrow located on nameplate.

3.6.2 Repair parts. Repair parts shall be marked in accordance with MIL-STD-130.

3.7 Workmanship. Workmanship shall be such as to assure a product free of burrs, rust, scratches, sharp edges, and chips.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the currentversion before use.

4.1.1 Materials. Contractor's inspection records shall be examined to determine conformance to 3.2.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all testing shall be conducted in ambient air temperature of 77°F ± 15°F, and not more than 90 percent relative humidity. Tests shall be conducted under general laboratory conditions acceptable to the Government.

4.3.1 Cap and rotor interference. After testing as specified in 4.6.1, distributor shall be operated for three hours at 300 rpm, and then three hours at 1800 rpm. Thereafter, distributor cap shall be removed and examined for interference with rotor. Satisfactory distributors, or repair parts, shall be reassembled and test sequence resumed.

4.4 Qualification inspection (see 6.3).

4.4.1 Distributor assemblies. A qualification sample of seven (7) distributor assemblies of each type and class to be qualified shall be furnished for qualification testing. Distributor assemblies shall be representative of distributor assemblies proposed to be furnished under contract. Qualification testing shall be conducted under Government surveillance by the contractor, or by an authorized testing facility, at a site approved by the Government. Inspection shall consist of examination for the defects specified in table IV and testing as specified in table III, in the order listed.

4.4.2 Authorized repair parts. Qualification sample shall consist of the following quantities of authorized repair parts:

Part	Number of samples
Cap	3
Cap cover	1
Coil	5
Contact set	2
Resistor	1
Rotor	3

Parts shall be representative of units proposed to be furnished to the Government. Qualification testing shall be conducted under Government surveillance by the contractor, or by an authorized testing facility, at a site approved by the Government. Qualification inspection shall consist of comparison examination of materials, dimensions, and finish with applicable drawings or MS Standards, or with another part having the same part number that has already been qualified. After comparison examination, part shall be assembled in a distributor assembly, examined for the defects specified in table IV, and tested as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

TABLE III. Order of qualification testing.

Specimen	Requirement	Test
Assembly 1	3.4.12.1	4.6.13 Resistor-suppressor
	3.4.2	4.6.2 Timing
	3.4.1	4.6.1 Voltage output
	3.4.13	4.6.14 High temperature operation
	3.4.1	4.6.1 Voltage output
	3.4.4	4.6.4 Endurance
	3.4.1	4.6.1 Voltage output
	3.4.3	4.6.3 Dielectric strength
	3.4.1	4.6.1 Voltage output
	3.4.13	4.6.14 High temperature operation
	3.4.1	4.6.1 Voltage output
	3.4.2	4.6.2 Timing
	3.4.12.1	4.6.13 Resistor-suppressor
2	3.4.7	4.6.7.1 Waterproofness
	3.4.3	4.6.3 Dielectric strength
	3.4.1	4.6.1 Voltage output
	3.4.14	4.6.15 Low temperature operation
	3.4.1	4.6.1 Voltage output
	3.4.16	4.6.17 Low temperature stall
	3.4.1	4.6.1 Voltage output
	3.4.3	4.6.3 Dielectric strength
	3.4.1	4.6.1 Voltage output
3	3.4.3	4.6.3 Dielectric strength
	3.4.1	4.6.1 Voltage output
	3.4.10.1	4.6.10.1 Vibration (pitch- or oil-filled coil)
	3.4.10.2	4.6.10.2 Vibration (molded or potted coil)
	3.4.1	4.6.1 Voltage output
	3.4.7	4.6.7.1 Waterproofness
	3.4.3	4.6.3 Dielectric strength
	3.4.1	4.6.1 Voltage output

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

TABLE III. Order of qualification testing. - Continued

Specimen	Requirement	Test	
Assembly	3.4.8	4.6.8	Corrosion resistance
	3.4.1	4.6.1	Voltage output
4	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
	3.4.15	4.6.16	High temperature stall
	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
5	3.4.9	4.6.9	Fungus resistance
	3.4.1	4.6.1	Voltage output
	3.4.7	4.6.7.1	Waterproofness
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
6	3.4.5	4.6.5	Polarity (Class 2 only)
	3.4.1	4.6.1	Voltage output
	3.4.6	4.6.6	Transient voltage (Class 2 only)
	3.4.1	4.6.1	Voltage output
	3.4.12	4.6.12	Electromagnetic compatibility
7	3.4.11	4.6.11	Shock resistance
	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
Cap 1	3.4.12.1	4.6.13	Resistor-suppressor
	3.4.13	4.6.14	High temperature operation
	3.4.1	4.6.1	Voltage output
	3.4.4	4.6.4	Endurance
	3.4.1	4.6.1	Voltage output
	3.4.12.1	4.6.13	Resistor-suppressor
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
Cap 2	3.4.7	4.6.7.1	Waterproofness
	3.4.14	4.6.15	Low temperature operation
3	3.4.10	4.6.10	Vibration
	3.4.1	4.6.1	Voltage output
	3.4.11	4.6.11	Shock resistance
	3.4.1	4.6.1	Voltage output
	3.4.12	4.6.12	Electromagnetic compatibility

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

TABLE III. Order of qualification testing. - Continued

Specimen	Requirement	Test	
Coil 1	3.4.1	4.6.1	Voltage output
	3.4.4	4.6.4	Endurance
	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
	3.4.13	4.6.14	High temperature operation
	3.4.1	4.6.1	Voltage output
2	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
	3.4.14	4.6.15	Low temperature operation
	3.4.10	4.6.10	Vibration
	3.4.1	4.6.1	Voltage output
3	3.4.11	4.6.11	Shock resistance
	3.4.1	4.6.1	Voltage output
	3.4.8	4.6.8	Corrosion resistance
	3.4.1	4.6.1	Voltage output
4	3.4.15	4.6.16	High temperature stall
	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
5	3.4.16	4.6.17	Low temperature stall
	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
Contact set 1 and 2	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
	3.4.4	4.6.4	Endurance
	3.4.1	4.6.1	Voltage output
Cap Cover 1	3.4.8	4.6.8	Corrosion resistance
	3.4.1	4.6.1	Voltage output
	3.4.10	4.6.10	Vibration
	3.4.1	4.6.1	Voltage output
	3.4.7	4.6.7.1	Waterproofness

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

TABLE III. Order of qualification testing. - Continued

Specimen	Requirement	Test	
Resistor			
1	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
	3.4.13	4.6.14	High temperature operation
	3.4.1	4.6.1	Voltage output
	3.4.14	4.6.15	Low temperature operation
	3.4.1	4.6.1	Voltage output
	3.4.4	4.6.4	Endurance
	3.4.1	4.6.1	Voltage output
Rotor			
1	3.4.12.1	4.6.13	Resistor-suppressor (if resistor is part of rotor)
	3.4.1	4.6.1	Voltage output
	3.4.13	4.6.14	High temperature operation
	3.4.1	4.6.1	Voltage output
	3.4.4	4.6.4	Endurance
	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
2	3.4.1	4.6.1	Voltage output
	3.4.10	4.6.10	Vibration
	3.4.1	4.6.1	Voltage output
	3.4.11	4.6.11	Shock resistance
	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
3	3.4.12	4.6.12	Electromagnetic compatibility

4.4.3 Retention of qualification. Certification shall be requested every two years from each manufacturer listed on the QPL to retain listing on the QPL. This certification shall be forwarded to the preparing activity and shall be signed by a responsible official of management, attesting that the listed product still meets the requirements of the current issue of the specification, is available from the listed plant, and can be produced under the same conditions as originally qualified; i.e., same process, materials, construction, design, manufacturer's part number or designation. Failure to provide certification shall be cause for removal from the QPL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. A lot shall consist of all components, or distributor assemblies of one type, class, and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.5.1.3 Sampling for acceptance testing. Samples for acceptance testing shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.5.2 Quality conformance examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined for conformance to the following acceptable quality levels (AQL's) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as follows:

TABLE IV. Classification of defects.

Categories	Defects	Method of inspection
Major		
101	Dimensions affecting interchangeability, not within tolerance (see 3.3 and 3.3.1)	Visual/ gage
102	Continuity of housing (see 3.3.2)	Visual
103	Coil-to-distributor lead (see 3.3.2.1)	Visual
104	Timing not adjustable (see 3.3.3)	Visual
105	Advance mechanism missing or inoperative (see 3.3.3.1)	Visual
106	Incorrect shaft speed (see 3.3.4)	Gage
107	Misalignment possible (see 3.3.4)	Visual
108 1/	Ventilation does not conform (see 3.3.5)	Visual
109 1/	Lubrication improper, or omitted (see 3.3.6)	Visual
110	Improper threaded parts (see 3.3.7)	Visual/ gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Categories	Defects	Method of inspection
Minor		
201	Dimensions not affecting interchangeability, not within tolerance (see 3.3)	Visual/ gage
202	Finish incomplete or improper (see 3.5)	Visual
203	Improper marking (see 3.6)	Visual
204	Faulty workmanship (see 3.7)	Visual

1/ Applies to distributor assembly.

4.5.2.3 Unclassified defects. All unclassified defects having no bearing on function, safety, interchangeability, or life, but which are considered departures from good workmanship, shall be noted in writing. Workmanship defects falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty day period, shall be added to the minor defects with no increase in AQL.

4.5.3 Classification of tests. Tests shall be classified as follows:

a. Acceptance tests (see 4.5.4).
b. Control tests (see 4.5.5).

4.5.4 Acceptance tests. All samples of distributor assemblies, selected in accordance with 4.5.1.3, shall be examined for the defects specified in table IV, and subjected to tests specified in table V, in order listed, using an AQL of 6.5 on the basis of percent defective.

TABLE V. Order of acceptance testing.

Requirement	Test	
3.4.3	4.6.3	Dielectric strength
3.4.1	4.6.1	Voltage output
3.4.7	4.6.7.2	Waterproofness
3.4.1	4.6.1	Voltage output
3.4.2	4.6.2	Timing
3.4.1	4.6.1	Voltage output

4.5.4.1 Authorized repair parts. Samples of repair parts (see 3.1.2.2) selected in accordance with 4.5.1.3, shall be examined for the defects specified in table IV, and subjected to tests specified in table VI in order listed, using an AQL of 6.5 on the basis of percent defective. Repair parts shall be substituted (see 3.1.2.2) in distributor assemblies that have previously passed inspection specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

TABLE VI. Order of acceptance testing - repair parts.

Repair part	Requirement	Test	
Cap	3.4.12.1	4.6.13	Resistor-suppressor
	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
	3.4.12.1	4.6.13	Resistor-suppressor
Coil	3.4.1	4.6.1	Voltage output
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
Cover	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
	3.4.7	4.6.7.2	Waterproofness
	3.4.1	4.6.1	Voltage output
Point set	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
Resistor	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output
Rotor	3.4.12.1	4.6.13	Resistor-suppressor (if resistor is part of rotor)
	3.4.3	4.6.3	Dielectric strength
	3.4.1	4.6.1	Voltage output

4.5.5 Control tests. Control test samples shall be identified as to production period and shall undergo tests specified in table VII, in order listed. Repair parts shall be substituted (see 3.1.2.2) in distributor assemblies that have previously passed inspection specified herein.

4.5.5.1 Sampling - control testing. Samples for control testing shall be selected from production lots which have passed quality conformance examination specified in 4.5.2. Control test samples shall be selected at the rate of one out of each 200 units produced, except that not less than one, and not more than two, shall be selected and tested in any 30 day period. Samples selected shall not have previously been subjected to acceptance testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

TABLE VII. Order of control testing.

Requirement	Test	
3.4.1	4.6.1	Voltage output
3.4.2	4.6.2	Timing
3.4.1	4.6.1	Voltage output
3.4.3	4.6.3	Dielectric strength
3.4.1	4.6.1	Voltage output
3.4.7	4.6.7.1 1/	Waterproofness

1/ Coil, point set, resistor, or rotor, when being tested as repair parts, need not be subjected to waterproofness test.

4.5.5.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.6 Conformance verification.

4.6.1 Voltage output test. To determine conformance to 3.4.1, distributor assembly shall be electrically connected and mechanically mounted in apparatus to simulate operation as in intended use. Each secondary lead shall be shunted through load capacitance of 150 ± 10 picofarads (pF). One secondary lead shall be connected to peak voltage measuring device. Remaining secondary leads shall be connected to spark gap as shown on drawing 7727531. Spark gap shall be three millimeters. Distributor assembly shall be operated at each input speed and voltage specified in table II. Distributor assembly shall be operated for three minutes before taking each measurement. Measurement of output voltage shall be taken at each lead, and recorded. Lowest voltage measured at any secondary lead shall be considered voltage output at that input speed and voltage. Thereafter, total capacitance of each secondary lead shall be adjusted to 150 ± 10 pF with resistance ungrounded. With 250,000 ohms ± 25,000 ohms shunted across secondary lead, and with 30 Vdc and 1800 rpm input, output voltage shall be measured and recorded at each secondary lead. Lowest reading shall be considered voltage output for distributor assembly (see 4.3.1).

4.6.1.1 Low input voltage test. To determine conformance to 3.4.1.1, class 2 distributor, assembled, mounted, and driven as in intended use, shall be subjected to correctly polarized input voltages between zero and 18 Vdc.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

4.6.1.2 High input voltage test. To determine conformance to 3.4.1.2, class 2 distributor, assembled, mounted, and driven as in intended use, shall be subjected to correctly polarized input voltages between 30 and 40 Vdc.

4.6.2 Timing test - advance and retard. To determine conformance to 3.4.2 and 3.4.2.1, distributor assembly shall be operated through range of speeds, and timing of spark shall be measured and recorded for comparison with requirements.

4.6.3 Dielectric strength test. To determine conformance to 3.4.3, each output lead of distributor shall be connected to spark gap, as shown on drawing 7727531, set at eight millimeters. No additional load shall be imposed on distributor. Input voltage shall be 32 Vdc ± 0.5 Vdc. Distributor shall be operated at 200 rpm for two minutes. Two output leads adjacent to coil shall then be connected to ground, remaining leads to eight millimeter gaps, and tests repeated for two minutes. Distributor shall then be examined for evidence of breakdown of insulation materials.

4.6.4 Endurance test. To determine conformance to 3.4.4, distributor assembly shall be electrically connected and mechanically mounted, as specified in 4.6.1, in ambient air temperature of 160°F ± 10°F. Spark gap, as shown on drawing 7727531, shall be set at three millimeters, and loaded with capacitance of 150 ± 10 pF. With input of 28.5 Vdc ± 0.5 Vdc, assembly shall be operated continuously through 300 cycles, if class 1, or 750 cycles, if class 2, of the following four phases (phases I-IV equal one cycle):

Phase	Time (minutes)	Distributor shaft speed (rpm)
I	30	300
II	30	800
III	30	1500
IV	30	2000

After each 100 hours (50 cycles) of operation as specified above, distributor shall undergo the following:

a. Observe sparking regularity for 10 minutes at each specified speed.
b. Observe output voltage at 300 rpm.
c. Observe output voltage at 1800 rpm.
d. Reset breaker points (class 1 only).
e. Apply trace of approved high temperature grease on breaker cam (class 1 only).
f. Apply one or two drops of approved light preservation oil on breaker lever pivot (class 1 only).

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

g. Apply three or four drops of approved light preservation oil on felt wick under rotor (class 1 only).

4.6.5 Polarity test - class 2 distributors. To determine conformance to 3.4.5, apply reverse polarity to the class 2 distributor, connected and operated as described in 4.6.1, except that no leads shall be connected to the peak voltage measuring device. A reverse polarity voltage of 30 volts shall be applied ten consecutive times for one minute, at four minute intervals.

4.6.6 Transient voltage characteristics test. To determine conformance to 3.4.6, the class 2 distributor shall be tested at 200°F and subjected to a properly polarized transient voltage of 240 Vdc above ground for 50 ± 5 milliseconds duration. The voltage shall be applied to the input terminals of the distributor ten consecutive times at one minute intervals.

4.6.7 Waterproofness tests.

4.6.7.1 Qualification and control test. To determine conformance to 3.4.7, distributor assembly shall be subjected to procedure specified in method 100, procedure 1 of MIL-STD-1184. Pretest performance shall be recorded.

4.6.7.1.1 Alternate qualification and control test. To determine conformance to 3.4.7, distributor assembly shall be tested as specified in 4.6.7.1, except that air pressurization and evacuation may be applied directly to interior of distributor that has access port for this purpose. Test chamber shall not be required. Distributor shall be submerged in open tank of water. Distributor, of qualified design that does not include access port, may not be altered for the purpose of using this method.

4.6.7.2 Acceptance test. To determine conformance to 3.4.7, samples selected for acceptance testing (see 4.5.1.3) shall be subjected to procedures specified in method 100, procedure 5 of MIL-STD-1184. Pretest performance shall be recorded.

4.6.7.2.1 Alternate acceptance test. To determine conformance to 3.4.7, distributor shall be tested as specified in 4.6.7.2, except that air pressurization may be applied directly to interior of distributor that has access port for this purpose. Test chamber shall not be required. Distributor shall be submerged in open tank of water. Distributor, of qualified design that does not include access port, may not be altered for the purpose of using this method.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

4.6.8 Corrosion resistance test. To determine conformance to 3.4.8, distributor assemblies shall be subjected to procedure specified in method 101 of MIL-STD-202 for a total of 200 hours. Subsequently, distributor shall be examined for conformance to 3.5.

4.6.9 Fungus resistance test. To determine conformance to 3.4.9, distributor assemblies shall be subjected to procedure specified in method B of MIL-F-13927 for a total of 90 days. Subsequently, distributor shall be examined for conformance to 3.5.

4.6.10 Vibration resistance test. To determine conformance to 3.4.10, distributor assembly shall be electrically connected and mechanically mounted in apparatus suitable to apply vibration in direction of each of three mutually perpendicular axes. Each distributor output lead shall be connected to spark gap, shown on drawing 7727531, set at three millimeters. With 28.5 Vdc input, and operating at 1200 rpm throughout test, distributor assembly shall be subjected to procedure specified in method 201 of MIL-STD-202.

4.6.10.1 With pitch- or oil-filled coil. Assembly with pitch- or oil-filled coil shall be subjected to 24 hours of specified vibration in each of three mutually perpendicular axes (total 72 hours).

4.6.10.2 With molded or potted coil. Assembly with molded or potted coil shall be tested in ambient air temperature of 180°F ± 5°F. Assemblies shall be subjected to 32 hours of specified vibration in each of three mutually perpendicular axes (total 96 hours). Testing shall be accomplished in two phases as follows:

Phase I	Distributor assembly shall be subjected to a simple harmonic motion having amplitude of 0.08 inches (0.16 inch maximum total excursion). Frequency shall be varied uniformly between 5 and 50 cycles per second (Hz). Frequency range shall be traversed in one minute, from 5 Hz to 50 Hz to 5 Hz.
Phase II	Amplitude shall be 0.02 inches (0.04 inch maximum total excursion). Frequency shall be varied uniformly between 50 Hz and 100 Hz. Frequency range shall be traversed in one minute, from 50 Hz to 100 Hz to 50 Hz.

Plane of vibration and phase shall be changed every eight hours during 96 hour procedure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

4.6.10.3 Coil mounting test. To determine conformance to 3.4.10.3, mount ignition coil in distributor base and secure with clamp assembly and screws. Assemble distributor in adapter 8754337 and mount in vibrator fixture so that vibration is in the plane of centerlines of the distributor shaft and coil. Apply 20g input vibration at the worst resonant frequency of the coil (between 60 and 150 Hz).

4.6.11 Shock resistant test. To determine conformance to 3.4.11, distributor assembly shall be subjected to procedure as specified in condition A, method 213, MIL-STD-202.

4.6.12 Electromagnetic interference characteristics test. To determine conformance to 3.4.12, distributor assemblies shall be subjected to applicable test procedures specified in MIL-STD-462.

4.6.13 Resistor-suppressor test. To determine conformance to 3.4.12.1, resistance values of resistor-suppressors shall be measured and recorded while undergoing short duration, high peak voltage pulses in accordance with the following:

Pulse repetition rate	4 ± 1 pulses per second
Pulse duration	100 microseconds or less
Pulse magnitude	5000 Vdc ± 250 Vdc
Pulse rise time	10 to 15 microseconds

4.6.14 High temperature operation test. To determine conformance to 3.4.13, distributor assembly shall be installed in a chamber in which ambient air temperature is controlled at 200°F ± 10°F.

4.6.15 Low temperature operation test. To determine conformance to 3.4.14, distributor temperature shall be stabilized at minus 65°F ± 5°F for 24 hours, removed from cold chamber and allowed to warm in air not more than 5 minutes, then operated as specified in 4.6.1, except that only one lead shall be checked and values obtained shall be within 25 percent of lowest value obtained in first output test.

4.6.16 High temperature stall test. To determine conformance to 3.4.15, distributor, with breaker apparatus at point of actuation, shall be placed in a test chamber at 160°F ± 5°F. After temperature has stabilized, test chamber shall be shut off, including air circulating fans, and chamber kept closed. Distributor shall immediately be energized from two 6TN batteries connected in series. Batteries shall be fully charged, in good condition, with electrolyte having corrected specific gravity of 1.280 ± .005 at 77°F. Distributor shall be energized in this manner for 6 hours.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

4.6.17 Low temperature stall test. To determine conformance to 3.4.16, distributor with breaker apparatus at point of actuation, shall be placed in a test chamber at minus 65°F ± 5°F for 24 hours. Distributor shall be energized from two 6TN batteries connected in series. Batteries shall be fully charged, in good condition, with electrolyte having corrected specific gravity of 1.280 ± .005 at 77°F. Distributor shall be energized in this manner for 6 hours.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Distributors furnished under this specification are intended for use in military motor vehicles, or other military equipment requiring waterproof ignition systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class (see 1.2).
c. Applicable drawing or MS standard (see 3.3).
d. Connection to the coil-to-distributor lead, when required (see 3.3.2.1).
e. Level of packaging, and level of packing (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement, and manufacturers are urged to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the parts covered by this specification. The activity responsible for the QPL is the Commanding General, US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, and information pertaining to qualification of products may be obtained from that activity.

6.4 Deletions. Appendix I has been deleted. Information formerly recorded there has been revised and incorporated in table I.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:21Z
Check the source to verify that this is the current version before use.

6.5 Definitions.

6.5.1 Point of actuation. Point of actuation is that point on circumference of distributor actuating device (see 6.5.2) at which current is actuated in secondary circuit.

6.5.2 Distributor shaft actuating device. Distributor actuating device is that portion of distributor that operates mechanism that actuates current in secondary circuit.

6.5.3 Breaker point design. Distributor assemblies in which secondary circuit is actuated by switch (contact set) operated by positive contact with distributor actuating device (cam), whether or not switch is in primary circuit, shall be referred to as "breaker point design" (class 1).

6.5.4 Breakerless design. Distributor assemblies in which secondary circuit actuation is induced by various non-mechanical means shall be referred to as "breakerless design" (class 2).

6.6 High peak voltage test generator. Equipment to produce high peak voltage necessary to calculate electrical resistance, as specified in 4.6.13, may be of any design producing required parameters. Information pertaining to approved equipment and test methods may be obtained from the US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, Michigan 48090.

6.7 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.8 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AT
Air Force - 99

Review activities:
Army - ME
DLA - CS

User activities:
Army - ER
Navy - MC

Preparing activity:
Army - AT

Project No. 2920-0365

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-D-13791

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:21Z
Check the source to verify that this is the current version before use.